Janeway: A Temporally Flexible Plan Executive

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Abstract

A robust executive is necessary to dispatch plans to robotic systems. These executives should be able to monitor plan execution to determine if they are going to be successful, and make adjustments to the plan to avoid failures. There are no existing executives that are both capable of flexible execution and are compatible with the standard interfaces for ICAPS planners and robotic systems. Existing executives that are compatible with these standard interfaces are unable to adapt the order or schedule of activities during execution. To close this gap, we present *Janeway*, a flexible plan executive capable of both relaxing grounded plans, and dynamically dispatching them. Janeway accomplishes this by combining causal link monitoring with a dynamic scheduler to achieve robustness with both respect to temporal constraints and state constraints. See https://youtu.be/sXDgGBwtE6M for additional details.

Introduction

In order for agents to execute plans, they require an executive to dispatch commands in real time. As online processes, executives have the information and ability to ensure that the plans are executed correctly. As the environment changes in real time, the executive monitors and makes adjustments to the plan.

A commonly used executive for executing PDDL plans is ROSPlan (Cashmore et al. 2015). ROSPlan provides a standardized interface to plug in PDDL planners, and dispatch commands over ROS nodes. However, in ROSPlan the core mechanism for ensuring robustness is grounded action plan monitoring. This means that ROSPlan stores the state of the world as a knowledge base, and before dispatching a new action, verifies that the knowledge base satisfies the preconditions of the action. If the conditions are not met, then the plan will fail and a replan is initiated. This ensures that no unsafe actions will be executed, however it assumes that the original ordering of actions must be fixed, and cannot be adjusted. Thus, ROSPlan will respond to unexpected changes by replanning, as opposed to potentially rescheduling the actions, while maintaining the same underlying activities. Replanning generally takes meaningfully

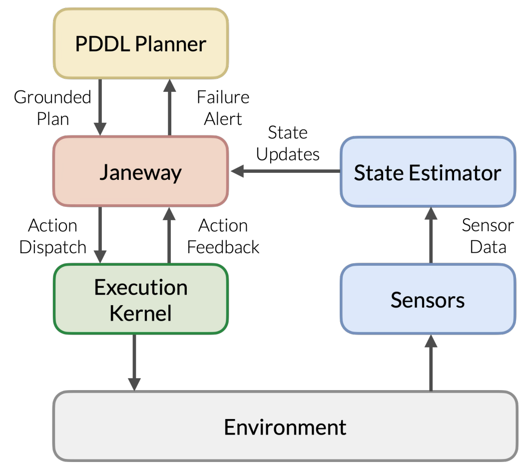

Figure 1: General overview of Janeway integration into an execution system. .

longer than rescheduling. Other executives can reorder actions during execution to avoid failure, or preemptively detect plan failures, however they do not provide standardized interfaces built for PDDL planner integration (Troesch et al. 2020; McGann et al. 2008).

To close this gap of flexible PDDL plan executives, we present Janeway. Janeway provides robustness in two dimensions: dynamic reordering/rescheduling, and early constraint violation detection. For the first capability, Janeway relaxes PDDL plans into a least-commitment ordering, and then uses the Continuous Real Time Execution algorithm to dynamically dispatch actions (Lee 2026). For the second capability, Janeway implements causal link monitoring (Levine and Williams 2014). As part of the plan relaxation process, Janeway identifies causal links, state conditions that must be maintained between two actions, that are necessary to ensure correct execution. Once a causal link is violated, Janeway halts execution and reports the failure to the user.

Janeway

Janeway consists of multiple modules which work together to relax, validate, execute, and monitor plans. We follow a similar level of abstraction as ROSPlan, where we interface

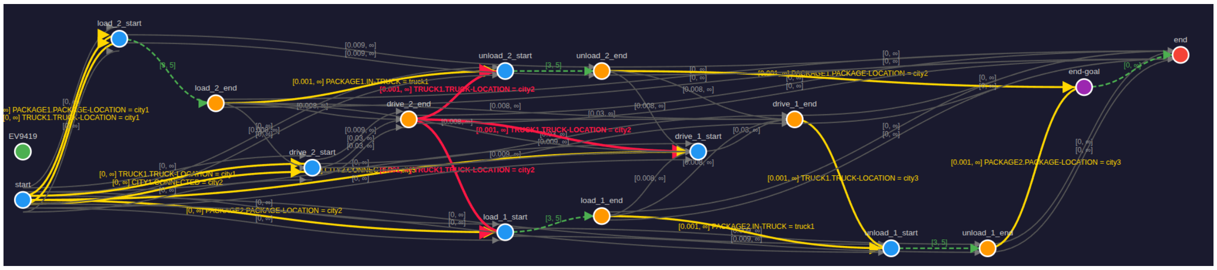

Figure 2: This is an example of a relaxed PDDL plan using Janeway's built-in visualization. Nodes represent the beginning and endings of actions, as well as the start and end of the plan. Yellow edges are causal links. Green edges correspond to the durations of actions. Grey edges are threats to causal links. Red edges are causal links violated during execution.

with the underlying agent or robot by dispatching high level activities as described in the plan, and listening for acknowledgments when the activities are completed and updates to the state of the world (See Figure 1).

Relaxation To enable both causal link monitoring and dynamic scheduling, Janeway relaxes input PDDL plans. The relaxation provides a least-commitment ordering of the activity plan, where ordering constraints are only added between activities that form a causal link, or between activities that would threaten a causal link. Causal link constraints are annotated with the corresponding precondition to facilitate future monitoring. Additionally, for temporal planning, Janeway supports actions with both controllable and uncontrollable durations. We provide a way to annotate the PDDL domain model with this information, so it can be incorporated into the relaxed plan.

Scheduling and Dispatching Janeway dynamically schedules and dispatches temporally controllable plans by extracting a Simple Temporal Network with Uncertainty (STNU) from the least commitment activity plan. STNUs encode temporal uncertainty by distinguishing between regular and contingent temporal constraints. Any event that occurs at the end of a contingent temporal constraint is marked as "uncontrollable", and therefore Janeway knows that it will not dispatch actions associated with that event, but rather listen for an external message that the event has occurred (Morris, Muscettola, and Vidal 2001). It then uses the Continuous Real Time Execution algorithm to dynamically schedule and dispatch activities, which is a variant of the Real Time Execution algorithm (Lee 2026; Tsamardinos, Muscettola, and Morris 1999).

Monitoring Janeway ensures safe action execution through causal link monitoring (Levine and Williams 2014). Using the causal links extracted during relaxation, Janeway knows if the start or end of an action is associated with either activating or deactivating a causal link. Incoming state updates are compared against the state constraints associated with all active causal links. If any of those constraints are violated, then the plan will fail, resulting in the halting of execution and a notification to the user. This allows Janeway to detect failures early, as preconditions are being validated throughout execution instead of just before the action is dispatched.

Simulated Execution Janeway comes with an internal simulator for executions which is referred to as the *Oracl*e. The Oracle both simulates external event acknowledgments for uncontrollable events, and state updates. When executions begins, the Oracle is sent the full plan that Janeway is about to execute so it knows which events will trigger a contingent temporal constraint, and by extension when it will need to simulate an uncontrollable event. Once an uncontrollable event is triggered, the oracle will randomly choose a time within the bounds of the contingent temporal constraint, and send Janeway a message at that time that the uncontrollable event has occurred. This allows users to test plan execution without writing their own event monitor to interface with Janeway.

Interfaces Janeway supports problems defined using PDDL 2.1, with or without durative actions (Fox and Long 2003). Janeway communicates over either HTTP or using ROS messages (Quigley et al. 2009). During execution, Janeway dispatches actions using their activity name from the PDDL domain, and a signifier to note if the action should be initiated or terminated. If an action is annotated as uncontrollable, Janeway will not dispatch commands to terminate the action but rather listen for an acknowledgment that the action has been completed. Additionally, Janeway provides a separate HTTP hook/ROS topic to report state updates. Janeway assumes that state updates are given in the form of assignments to predicates as defined in the PDDL domain. Users can view the relaxed version of their PDDL plans and monitor their execution via the web-based visualization tool, which shows both when activities occur as well as what state updates Janeway receives.

Conclusion

In conclusion, Janeway is a flexible, easy-to-use PDDL plan executive. It is the only executive that brings dynamic scheduling and causal link monitoring to PDDL planners. We have tested this system both against using our own internal execution simulator and through interfacing with ROS modules interacting with the Gazebo simulator.

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